

Eshani Patel

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EDUCATION

Carnegie Mellon University | Pittsburgh, PA Aug. 2026 – Dec. 2027
Master of Science in Machine Learning
California Institute of Technology | Pasadena, CA Sep. 2021 – Mar. 2025
Bachelor of Science in Computer Science, Minor: Information & Data Science, GPA: 4.1/4.3

WORK EXPERIENCE

Software Engineer - Systems & Infrastructure, LinkedIn | Sunnyvale, CA Aug. 2025 – Sep. 2026
Building fault-tolerant, exabyte-scale, multi-tenant, multi-medium distributed foundational storage platform.

- Rewrote stack with a reactive concurrency model for high throughput and latency I/O, enabling Adaptive Rate Limiting for graceful node degradation under overload; **reducing CPU consumption by 15%** and ensuring predictable performance under overload.
- Extended the I/O stack with pluggable encoding support (Replication and Erasure Coding), enabling cost-optimized storage service tiers and extensible library.
- Implemented quorum writes, **reducing write latency by 40%** with negligible durability impact; enabled lower-cost node placement without write latency regression.
- Built a chaos-driven fault injection pipeline; **detected 5 critical bugs** pre-launch, uncovered missing alert notifications via FireDrill testing, and serves as release certification pipeline.

Software Engineering Intern, LinkedIn | Sunnyvale, CA June 2024 – Sep. 2024

- Developed and deployed XGBoost/Random Forest models to estimate compute usage for on-demand reports, cutting infrastructure costs on AB Testing platform.
- Built model integration with Java microservice API and ETL pipelines with Apache Spark (Scala) to process **1M+** usage records.

Software Engineering Intern, Rakuten | San Mateo, CA June 2023 – Aug. 2023

- Built product history logging system in Java for Catalog Platform, enhancing data traceability and supporting analytics at scale.
- Utilized Google Cloud and Apache Beam for ETL of product information in data pipeline of **1M+** records.

Computer Science Head Teaching Assistant & Course Instructor, Caltech | Pasadena, CA Jan. 2023 – Dec. 2024

- Led 20-person TA staff for Software Design (**120+** students); co-taught interview prep course (**60+** students), with alumni placed at Google, Meta, Amazon, Microsoft, and Apple.

RESEARCH EXPERIENCE

ML Researcher, Yue Group, Caltech | Pasadena, CA Mar. 2025 – Aug. 2026

- Extended Population Transformer, a self-supervised iEEG time-series foundation model, to variable-length sequences via data-augmented pretraining, facilitating population-level representation learning.
- Applied AutoResearch agentic loop (Recursive Self Improvement) for automated iEEG foundation model pretraining tuning.
- **Reduced training pipeline memory footprint by 30%**, enabling training on lower-spec servers.

Undergraduate Researcher, Computational Cameras Group, Caltech | Pasadena, CA Jan. 2024 – Mar. 2025

- Developed multimodal conditional diffusion model for dark matter field reconstruction from astronomical simulations; integrated two novel observational modalities and demonstrated robustness across SNR levels.
- Investigated scalability of Score-based Image Generation Models using StyleGAN for high-res image denoising and reconstruction.

Machine Learning Intern, NASA Jet Propulsion Laboratory | Pasadena, CA June 2022 – Aug. 2022

- Developed Random Forest models to predict atmospheric ozone levels and quantify physical model bias; improved accuracy **15%** via Google Earth Engine topographic feature engineering.

PUBLICATIONS

Patel, E., Yue, Y., Chau, G. Learning Time-Scale Invariant Population-Level Neural Representations. *Conference on Neural Information Processing Systems (NeurIPS) 2025: Workshop on Foundation Models for the Brain and Body*. [arxiv]

Patel, E., Marchetti, Y., Miyazaki, K., et al. Towards Understanding the Sub-scale Drivers of Surface Ozone with ML-Based Feature Permutation Importance Analysis. *American Meteorological Society Conference 2023*. [link]

SELECTED PROJECTS

Eikonal Tomography with Fourier Neural Operators: Applied FNOs for seismic wave travel-time estimation, replacing PDE solvers; scalable training on 100GB datasets using GPU-accelerated HPC clusters.

ReAct-Based Semantic Wikipedia Scraper: Agentic AI framework with Chain-of-Thought reasoning and semantic Wikipedia search; higher retrieval precision over standard RAG baselines.

SKILLS

ML Expertise: Self-supervised & Representation Learning, Generative Modeling (Diffusion, GANs, VAEs), Foundation Models, Transformer Architectures, Time-series Modeling

ML Frameworks: PyTorch, Scikit-learn, HuggingFace Transformers, NumPy, Pandas, Weights & Biases

Systems & Infrastructure: Docker, Kubernetes, REST APIs, gRPC, Apache Spark, Apache Beam, GCP, Git, Linux, Slurm

Programming: Python, Rust, Java, C, C++, Scala, SQL

Courses: Machine Learning, Deep Learning Systems, AI/ML Learning Systems, Large Language & Vision Models, LLM Agents, Distributed Computing, Statistical Inference, Linear Algebra, Probability/Statistics, Databases, Networks